Task.java:->

```
package com.example.demo;

public class Task implements Runnable{
    private String taskName;
    public Task(String taskName)
    {
        this.taskName=taskName;
    }


    @Override
    public void run() {
        System.out.println("Executing task:"+taskName+"by
thread:"+Thread.currentThread().getName());
        try
        {
            Thread.sleep(2000L);
        } catch (InterruptedException e) {
            throw new RuntimeException(e);
        }
        System.out.println(taskName+" is Completed.");
    }
}
```

ExecutorServiceExample.java:->

```java
package com.example.demo;

import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.concurrent.TimeUnit;

public class ExecutorServiceExample {
    public static void main(String[] args) {
        ExecutorService executorService= Executors.newFixedThreadPool(3);
        for(int i=0;i<6;i++)
        {
            Task task=new Task("Task "+i);
            executorService.submit(task);
        }
        System.out.println("All tasks submitted");
        try
        {
            if(!executorService.awaitTermination(60, TimeUnit.SECONDS))
            {
                executorService.shutdown();

            }
        } catch (InterruptedException e) {
            executorService.shutdown();
        }
        System.out.println("Executor service shutdown");

    }
}
```

ScheduledExecutorServiceExample.java:->

```java
package com.example.demo;

import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.concurrent.ScheduledExecutorService;
import java.util.concurrent.TimeUnit;

public class ScheduledExecutorServiceExample {
    public static void main(String[] args) {
        ScheduledExecutorService scheduledExecutorService=
Executors.newScheduledThreadPool(3);
        ExecutorService singleThreadPool=Executors.newSingleThreadExecutor();
        ExecutorService fixedThreadPool=Executors.newFixedThreadPool(4);
        ExecutorService cachedThreadPool=Executors.newCachedThreadPool();
        ExecutorService
scheduledThreadPool=Executors.newScheduledThreadPool(4);
        ExecutorService
singleScheduledThreadPool=Executors.newSingleThreadScheduledExecutor();

        for(int i=0;i<5;i++)
        {
            Task task=new Task("Task "+i);
            scheduledExecutorService.schedule(task,i*2, TimeUnit.SECONDS);
        }
        scheduledExecutorService.scheduleAtFixedRate(
                new Task("Fixed rate task")
                ,1
                ,5
                ,TimeUnit.SECONDS

        );

        scheduledExecutorService.scheduleWithFixedDelay(
                new Task("fixed delay task")
                ,2
                ,5
                ,TimeUnit.SECONDS
        );

        try
        {
            Thread.sleep(20000);

        } catch (InterruptedException e) {
            e.printStackTrace();
        }
        scheduledExecutorService.shutdown();
        System.out.println("scheduledExecutorService shutdown");

    }
}
```